

BAYESIAN ANALYSIS OF BIVARIATE INTERVAL CENSORED SURVIVAL DATA

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Summary: In the analysis of survival data, lifetimes may be defined by intervals rather than exact lifetimes. Furthermore, if there are several failure types a subject's response can be described by a number of intervals, where the failure times of the different failure types can be correlated. Covariates are introduced by means of the proportional hazards model. Previous authors have used the working assumption of independence between failure types, conditional on the covariates, so that correlation is only introduced by the common effect of the covariates on the failure types. In this paper, in a nonparametric setting, two methods of introducing correlation are proposed to allow for conditional dependence in the case of bivariate failure times, which allows for the possibility that the covariates have different effects on the two failure times. Independent Dirichlet priors on the interval probabilities are assumed and posterior measures are obtained by MCMC methods. It is found in a simulation study that failure to account for conditional dependence can result in severely biased inferences.

1. Introduction

The time to an event (failure) cannot always be ascertained exactly. Instead, observations may be grouped together according to whether they fall within an interval $[t_{j-1}, t_j)$, where $t_0 < t_1 < \dots < t_{r+1}$, $t_0 = 0$ and $t_{r+1} = \infty$.

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This scenario arises, for example, if an experimental unit is regularly clinically observed for occurrence of an event at predetermined intervals. Should compliance be uniform for all units, then observed data will be in the form of equally spaced, non-overlapping time intervals. Prentice and Gloeckler (1978) termed this grouped survival data. This kind of data frequently arises in biomedical sciences and in long-term clinical trials such as chronic disease and AIDS studies. There is a large body of work on univariate interval censored data, first described by Turnbull (1976). Most papers deal with the analysis of the proportional hazards model for grouped data. See, for example, Lawless (1982), Finkelstein (1986), Satten (1996), Goggins, et al. (1998) and Hudgens, et al. (2001). Papers that deal with multivariate interval censored failure time data are Guo and Lin (1994) and Goggins and Finkelstein (2000), who generalized the work of Guo and Lin so that intervals of individual subjects can overlap and are of varying lengths. Both the above papers use Cox's proportional hazards model to introduce covariates and derive the maximum likelihood estimators. The assumption of conditional independence is made, which means that correlation between failure types is only the result of the common effect of the covariates. In many applications this assumption is not realistic due to the effect of unexplained common factors that cause conditional or inherent dependence. For example, the multiple failure times of the same person, or the disease incidence of related individuals, have many common factors not explained by the covariates. Prentice and Cai (1992) use the covariance between counting process martingales to characterize the inherent dependence between two failure types in the case of right-censored failure time data. An advantage of accounting for inherent dependence is that the effect of the covariates on the various failure times need not be common,

and this assumption may be a necessary in many applications. One way of dealing with multivariate failure times is through frailty models, which allow for a common frailty variable for all events of a single subject or cluster of subjects. The frailty model is especially suitable when a subject can experience a random number of repeated failures. The Bayesian treatment of frailty models can be found in Clayton (1991), Sinha (1993), Sinha and Dey (1997), Sinha, Chen and Ghosh (1999) as well as the book by Ibrahim, Chen and Sinha (2001). In this paper we consider strictly paired failure times, and propose two bivariate nonparametric Bayesian models, incorporating conditional dependence. For the first model we define probabilities for the second failure type, conditional on the first failure, while for the second model we use the bivariate survival function defined by Clayton (1978). Although the Clayton model is well known, we believe that the incorporation of interval censored data in a nonparametric setting, combined with a Bayesian analysis, has not been considered before.

2. Preliminaries

For interval-censored data, the probability of an event occurring in the time interval $I_j = (t_{j-1}, t_j]$, is given by

$$P(T \in I_j) = \theta_j = S(t_{j-1}) - S(t_j), \quad (2.1)$$

where $S(\cdot)$ is the survival function. Of fundamental importance is the baseline conditional probability of survival beyond interval $(t_{j-1}, t_j]$, given survival to the interval;

$$P_j = P[T > t_j | T > t_{j-1}] = \frac{P(T > t_j)}{P(T > t_{j-1})}. \quad (2.2)$$

The unconditional probability of failure in interval I_j can then be written as

$$\theta_j = (1 - P_j) \prod_{s=1}^{j-1} P_s, \quad j = 1, 2, \dots, r, \quad (2.3)$$

and

$$S(t_j) = \prod_{s=1}^j P_s. \quad (2.4)$$

Let $\mathbf{z}(p \times 1)$ be a vector of fixed covariates, and let an experimental unit be subjected to M failure types, each having survival space subdivided into r intervals ($j = 1, 2, \dots, r$), denoting the checking times. The i th subject's failure is described by a vector $T_i = \{t_{i1}, t_{i2}, \dots, t_{iM}\}$ specifying the intervals in which each failure type occurs. Then P_{mj} is the conditional probability that a unit experiences the m th failure type after the j th interval. Under the proportional hazards model, the probability that a unit with covariates $\mathbf{z}$ survives beyond interval I_{mj} for the m th failure type is then

$$S(t_{mj}) = \left(\prod_{s=1}^j P_{ms} \right)^{\exp(\beta \mathbf{z})}, \quad (2.5)$$

where $\beta(1 \times p)$ is the vector of unknown regression coefficients. This model is called the grouped Cox model since it is derived as a grouped version of the continuous proportional hazards model. We use the log-log transformation to eliminate range restrictions,

$$P_{mj} = \exp(-\exp(\gamma_{mj})), \quad (2.6)$$

where γ_{mj} can be viewed as the baseline interval effect. Define a dichotomous random variable α_{im} so that

$$\alpha_{im} = \begin{cases} 1 & \text{if the } i\text{th item's } m\text{th event is a failure} \\ 0 & \text{if the } i\text{th item's } m\text{th event is censored.} \end{cases}$$

Then the contribution to the likelihood of the i th observation, whose m th failure type falls in the j th interval, is

$$\begin{aligned}
 L_{imj}(\beta, \gamma) &= (P_{mj}(\mathbf{z}_i))^{1-\alpha_{im}} \\
 &\quad \times (1 - P_{mj}(\mathbf{z}_i))^{\alpha_{im}} \prod_{s=1}^{j-1} P_{ms}(\mathbf{z}_i) \\
 &= (\exp(-\exp(\gamma_{mj} + \beta \mathbf{z}_i)))^{1-\alpha_{im}} \\
 &\quad \times (1 - \exp(-\exp(\gamma_{mj} + \beta \mathbf{z}_i)))^{\alpha_{im}} \\
 &\quad \times \prod_{s=1}^{j-1} \exp(-\exp(\gamma_{ms} + \beta \mathbf{z}_i)) \quad (2.7)
 \end{aligned}$$

where

$$P_{mj}(\mathbf{z}_i) = P[T_{im} > t_j | T_{im} > t_{j-1}, \mathbf{Z} = \mathbf{z}_i]. \quad (2.8)$$

Under the assumption of conditional independence (Guo & Lin, 1994, Goggins & Finkelstein, 2000), the full likelihood is then

$$L(\beta, \gamma) = \prod_{i=1}^n \prod_{m=1}^M \sum_{j=1}^{r+1} \phi_{imj} L_{imj}(\beta, \gamma), \quad (2.9)$$

where

$$\phi_{imj} = \begin{cases} 1 & \text{if the } i\text{th item's } m\text{th event (failure or censoring) occurs in } I_j \\ 0 & \text{otherwise.} \end{cases}$$

In the case of missing observations, the interval of failure, say $(L_i, R_i]$, may cover several standard intervals. Then the indicator $\phi_{imj} = 1$ if $I_j \subset (L_i, R_i]$ and zero otherwise. The above model assumes that dependence is only created through the covariates, and that the covariates have a common effect on all M failure types. The assumption would be unrealistic in many applications.

3. Conditional bivariate model

Consider $M = 2$ failure types and define $P_{1j}(z_i)$ as in (2.8) for the first failure type. Then define $P_{2qj}(z_i)$ as the probability that the second failure falls in its q th interval, given that the observed value of the first failure type lies in its j th interval.

$$P_{2qj}(z_i) = P[T_{2i} > t_q | T_{2i} > t_{q-1}, t_{j-1} < T_{1i} \leq t_j, \mathbf{Z} = z_i], \quad (3.1)$$

where $q = 1, \dots, r$; $j = 1, \dots, r+1$; $t_{r+1} = \infty$. In this way inherent correlation between failure types is introduced and the effect of the covariates need not be the same for both failure types. For simplicity of notation the number of intervals for both variables is taken equal to r , but the generalisation is straightforward. Then

$$\begin{aligned} P_{ij}(z_i) &= \exp(-\exp(\gamma_{1j} + \beta_1 z_i)), \quad j=1, 2, \dots, r, \\ \text{and} \\ P_{2qj}(z_i) &= \exp(-\exp(\gamma_{2qj} + \beta_2 z_i)), \quad q=1, 2, \dots, r; j=1, 2, \dots, r+1. \end{aligned}$$

The contribution to the likelihood of the i th item, whose events took place in the j th and q th intervals, is

$$\begin{aligned} L_i(\beta, \gamma) &= \left[(1-P_{1j}(z_i)) (1-P_{2qj}(z_i)) \prod_{s=1}^{j-1} P_{1s}(z_i) \prod_{s=1}^{q-1} P_{2sj}(z_i) \right]^{\alpha_{i1j}\alpha_{i2q}} \\ &\times \left[(1-P_{1j}(z_i)) \prod_{s=1}^{j-1} P_{1s}(z_i) \prod_{s=1}^q P_{2sj}(z_i) \right]^{\alpha_{i1j}(1-\alpha_{i2q})} \\ &\times \sum_{s=j+1}^{r+1} \left[(1-P_{1s}(z_i)) (1-P_{2qs}(z_i)) \prod_{\ell=1}^{s-1} P_{1\ell}(z_i) \prod_{\ell=1}^{q-1} P_{2\ell s}(z_i) \right]^{(1-\alpha_{i1j})\alpha_{i2q}} \\ &\times \sum_{s=j+1}^{r+1} \left[(1-P_{1s}(z_i)) \prod_{\ell=1}^{s-1} P_{1\ell}(z_i) \prod_{\ell=1}^q P_{2\ell s}(z_i) \right]^{(1-\alpha_{i1j})(1-\alpha_{i2q})} \end{aligned} \quad (3.2)$$

where $P_{1, \tau+1}(\mathbf{z}_i) = 0$. The full likelihood is then

$$L(\beta, \gamma) = \prod_{i=1}^n \sum_{j=1}^{\tau+1} \sum_{q=1}^{\tau+1} \omega_{ijq} L_i(\beta, \gamma), \quad (3.3)$$

where $\omega_{ijq} = 1$ if the first event occurs in the j th interval and the second event occurs in the q th interval, and zero otherwise. The derivation of one of the terms in (3.2) is given in the appendix. If T_1 and T_2 are conditionally independent, then $P_{2qj} = P_{2q}$ for all j and (3.2) reduces to

$$\begin{aligned} L_i(\beta, \gamma) &= \left[(1 - P_{1j}(\mathbf{z}_i)) \prod_{s=1}^{j-1} P_{1s}(\mathbf{z}_i) \right]^{\alpha_{11j}} \left[\prod_{s=1}^j P_{1s}(\mathbf{z}_i) \right]^{(1-\alpha_{11j})} \\ &\times \left[(1 - P_{2q}(\mathbf{z}_i)) \prod_{s=1}^{q-1} P_{2s}(\mathbf{z}_i) \right]^{\alpha_{12q}} \left[\prod_{s=1}^q P_{2s}(\mathbf{z}_i) \right]^{(1-\alpha_{12q})} \end{aligned} \quad (3.4)$$

4. The Clayton model

Clayton (1978) proposed a model for association in bivariate survival functions for the analysis of familial tendency in disease incidence. Comparisons between parent and offspring or between siblings for possible relationships between disease incidences in related individuals require a model for dependent events. This model is also suitable for the analysis of bivariate failure types in a single individual. In general, the bivariate Clayton survival function can be written as

$$S(t_{1j}, t_{2q}) = \left[S_1(t_{1j})^{-1/\theta} + S_2(t_{2q})^{-1/\theta} \right]^{-\theta}, \quad (4.1)$$

where $S_m(\cdot)$ is the marginal survival function. The parameter of association is θ , and $\theta \rightarrow 0$ for perfect correlation while $\theta \rightarrow \infty$ for independence. In

(4.1) the individual survival functions are now given by

$$S_m(t_{mj}) = \prod_{s=1}^j P_{ms}(\mathbf{z}_i) = \prod_{s=1}^j \exp(-\exp(\gamma_{ms} + \beta_m \mathbf{z}_i)), \quad m = 1, 2. \quad (4.2)$$

Then the contribution to the likelihood of the i th item, whose events took place in the j th and q th intervals, is

$$L_i(\beta, \gamma) = (2\alpha_{i1}-1)(2\alpha_{i2}-1)S(t_{1j}, t_{2q}) - (2\alpha_{i1}-1)\alpha_{i2}S(t_{1j}, t_{2, q-1}) \\ - \alpha_{i1}(2\alpha_{i2}-1)S(t_{1, j-1}, t_{2q}) + \alpha_{i1}\alpha_{i2}S(t_{1, j-1}, t_{2, q-1}), \quad (4.3)$$

where α_{im} is defined in section 3. The full likelihood is then again as in (3.3).

5. Prior distributions

We assume uniform priors, $\pi(\beta_1, \beta_2) \propto 1$, for the regression coefficients β_1 and β_2 . For the conditional model of section 3 we have interval parameters γ_{1j} , $j = 1, 2, \dots, r$, and γ_{2qj} , $q = 1, 2, \dots, r$, $j = 1, 2, \dots, r+1$, while for the Clayton model we have γ_{mj} , $m = 1, 2$ and $j = 1, 2, \dots, r$. To obtain non-informative priors for the γ 's, we start by assuming independent and identical Dirichlet priors on all the interval probabilities for both failure types. So if $(\theta_{m1}, \theta_{m2}, \dots, \theta_{mr})$ are the interval probabilities for the m th failure type, then

$$\pi(\theta_{m1}, \theta_{m2}, \dots, \theta_{mr}) \propto \prod_{j=1}^r \theta_{mj}^{\delta_j-1} \left(1 - \sum_{k=1}^r \theta_k\right)^{\delta_{r+1}-1}. \quad (5.1)$$

The conditional probabilities can be written in terms of the interval probabilities as

$$P_{mj} = \frac{1 - \sum_{k=1}^j \theta_{mk}}{1 - \sum_{k=1}^{j-1} \theta_{mk}}$$

which, after transformation, results in independent Beta $\left(\sum_{k=s+1}^{r+1} \delta_k, \delta_s \right)$ priors on all P_{1s} and P_{2sj} , $s = 1, 2, \dots, r$. For the link function, $\gamma_{mj} = \log(-\log(P_{mj}))$ we then obtain the prior

$$\pi(\gamma_{mj}) \propto \exp \left(\gamma_{mj} - \exp \left(\gamma_{mj} \sum_{k=j+1}^{r+1} \delta_k \right) (1 - \exp(-\exp(\gamma_{mj}))) \right)^{\delta_j - 1}. \quad (5.2)$$

In general, to represent vague prior information, we will take all δ_{1j} and δ_{2qj} equal to δ , where δ is small, $0 < \delta \leq 1$. For intervals of unequal lengths, the δ_{mj} s could be chosen proportional to the interval lengths. Note that the $(r+1)$ st interval is open on the right. If $\delta = 1$, equation (5.2) reduces to the extreme value distribution;

$$\pi(\gamma_{mj}) = (r+1-j) \exp(\gamma_{mj} - (r+1-j) \exp(\gamma_{mj})), \quad (5.3)$$

and the Jeffreys prior ($\delta = 0.5$) is given by

$$\begin{aligned} \pi(\gamma_{mj}) &= \frac{1}{B((r+1-j)/2, 1/2)} \exp(\gamma_{mj} - 1/2(r+1-j) \exp(\gamma_{mj})) \\ &\quad \times (1 - \exp(-\exp(\gamma_{mj})))^{1/2}. \end{aligned} \quad (5.4)$$

For the Clayton model we assume for θ the vague prior $\pi(\theta) \propto \theta^{-1}$. Because of the improper prior on θ , it is unclear whether the posterior will be proper. The problem is only in the upper tail of the distribution because the posterior approaches zero as $\theta \rightarrow 0$. The problem can be overcome by putting an upper bound on θ (see Figure 2). Where the conditional model has $2(p+r) + r^2$

parameters, the Clayton model has only $2(\tau + p) + 1$ parameters but assumes a more rigid structure in the association between variables.

6. Illustration

To illustrate the models, we simulated 200 observations from the bivariate Clayton model in (4.1) with Weibull marginals so that $S_m(t) = \exp\left(-(\alpha_m t)^{\phi_m}\right)$, $m = 1, 2$, and with measure of association $\theta = 0.6$. The Weibull parameters are $\alpha_1 = \alpha_2 = 0.5$ and $\phi_1 = 2$, $\phi_2 = 4$. One covariate was used, with $\beta_1 = -0.2$ and $\beta_2 = 0.5$, so that the covariate has different effects on the two variables (see the Appendix). The inherent (baseline) correlation between the two failure times is about 0.65, but this is reduced to 0.49 in this sample because of the conflicting effects of the covariate on the two failure times. The sample spaces for both variables were divided into seven intervals, the last of which is an open interval containing censored observations.

Table 1. Bayes estimates for β_1, β_2, θ and the conditional probabilities of the first failure, $P_{1j} = [T_1 > t_{1j} | T_1 > t_{1,j-1}]$, $j = 1, \dots, 6$

Parameter	True value	Clayton model	Conditional model
β_1	-0.2	-0.199 (-0.33; -0.06)	-0.217 (-0.38; -0.04)
β_2	0.5	0.606 (0.44; 0.79)	0.710 (0.48; 0.96)
θ	0.6	0.723 (0.44; 1.01)	
P_{11}	0.939	0.960	0.969
P_{12}	0.829	0.821	0.825
P_{13}	0.732	0.740	0.747
P_{14}	0.646	0.599	0.605
P_{15}	0.570	0.613	0.625
P_{16}	0.503	0.461	0.466

MCMC simulations were run using both models to see if these models can distinguish between the inherent relationship and the effect of the covariate by estimating the effect of the covariate on each of the failure times correctly. The model of Goggins and Finkelstein (2000), which assumes conditional independence, is not applicable here since a common value for β has to be assumed. The resulting Bayes estimators are given in Table 1. It is clear from Table 1 that both models distinguish quite well between the effects of inherent correlation, the interval effects and the effects of the covariates on the two failure types. If the conditional independence model is used, an estimate of β close to zero is obtained, which is the result of the opposing effects of the two covariates.

7. Application

Data from McGilchrist and Aisbett (1991), also analysed by Ibrahim, Chen and Sinha (2001), give the recurrence times to infection for kidney patients who use portable dialysis equipment. Two recurrence times are given for each patient. The second recurrence time is measured from the time the first infection has cleared up. The covariates are age and sex. Some observations are censored due to a break in treatment for other reasons.

Assuming a common effect of the covariates on both failure times, we have a single regression vector, $\beta = (\beta_1 \beta_2)$. Even though the exact failure times are known in this study, the times for both failures were grouped into one of five intervals for the application and illustration of our model: 0 – 30; 31 – 90; 91 – 180; 181 – 365; 366 – 600 days. There are 38 paired observations, and the conditional model has 37 parameters while the Clayton model has 13 parameters. Using the Metropolis-Hastings algorithm with block simulations, 20 000 sets of generated values were kept using both

models, and the Jeffreys prior (5.4) was used. Priors for γ_j , $j = 1, \dots, 5$, are shown in Figure 1.

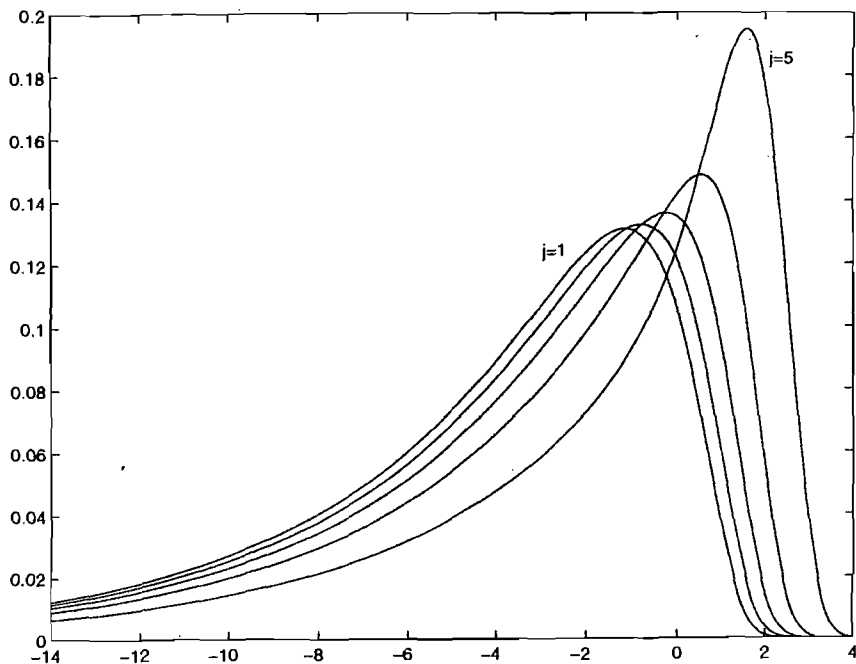


Figure 1. Prior distributions for γ_j , $j = 1, \dots, 5$

Figure 2 shows the posterior of θ in the case of the Clayton model. Table 2 shows some posterior measures. It is clear that age has little or no effect on failure times, while sex has a significant effect. The same conclusions were reached by McGilchrist and Aisbett (1991) using a frailty model.

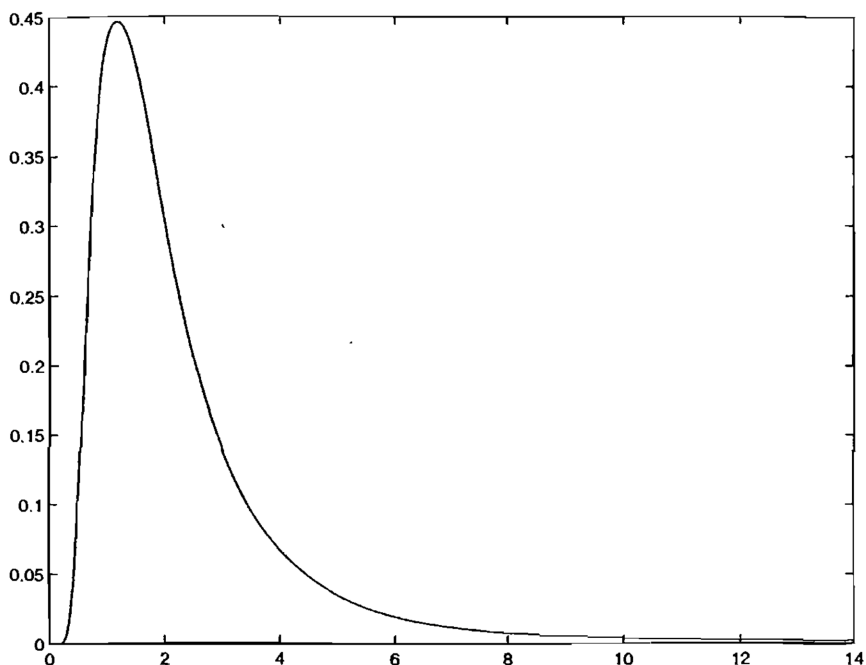


Figure 2. Posterior distribution of association parameter θ in the Clayton model

Table 2. Bayes estimates for β_1 , β_2 , θ and the conditional probabilities of the first failure, $P_{1j} = [T_1 > t_{1j} | T_1 > t_{1,j-1}]$, $j = 1, \dots, 5$ for the kidney data

Parameter	Clayton model	Conditional model
β_1 (age)	0.031 (-0.155; 0.213)	0.035 (-0.126; 0.220)
β_2 (sex)	1.040 (0.299; 1.739)	1.252 (0.541; 1.998)
θ	2.415 (0.401; 5.927)	
P_{11}	0.778	0.800
P_{12}	0.839	0.841
P_{13}	0.644	0.632
P_{14}	0.687	0.722
P_{15}	0.006	0.012

It is difficult to assess the strength of association between the first and second failures from the estimated value $\hat{\theta} = 2.4$ of the parameter θ in the Clayton model, but the effect can be seen by looking at the conditional survival functions. Figure 3 shows the estimated survival functions of the first failure for men and women aged 50. They are very similar in both models. Figure 4 shows for the same covariates the conditional survival functions, $S(t_2 | t_1 < 30)$, for the second failure given that the first failure was in the first interval (within 30 days). The differences between the unconditional survival function of T_2 and those of T_2 given different intervals for T_1 , are an indication of the effect of the inherent correlation. In this application there is only a mild effect.

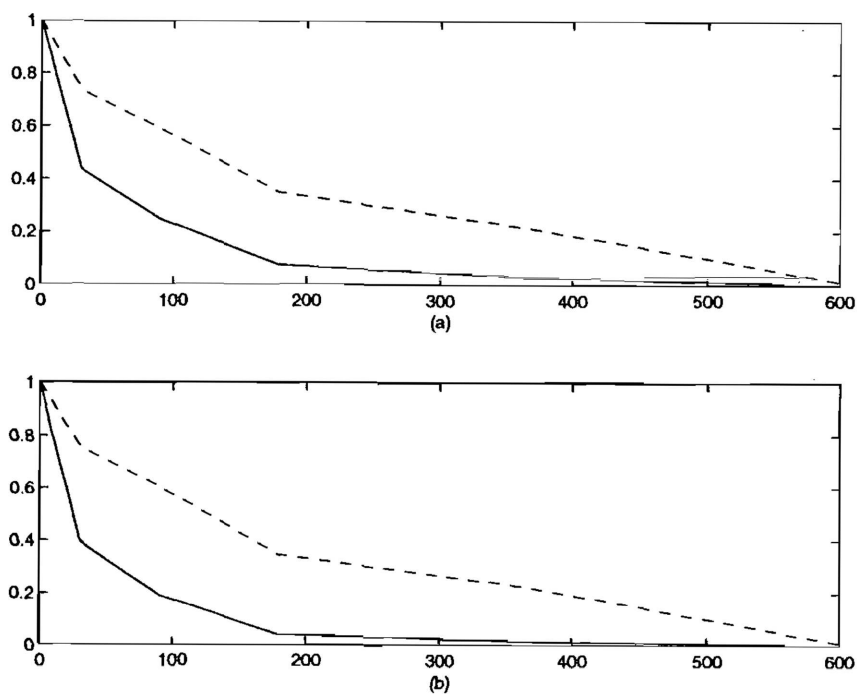


Figure 3. Estimated survival functions of first failure for men (solid) and women (dashed) at age 50, with (a) Clayton model and (b) Conditional model

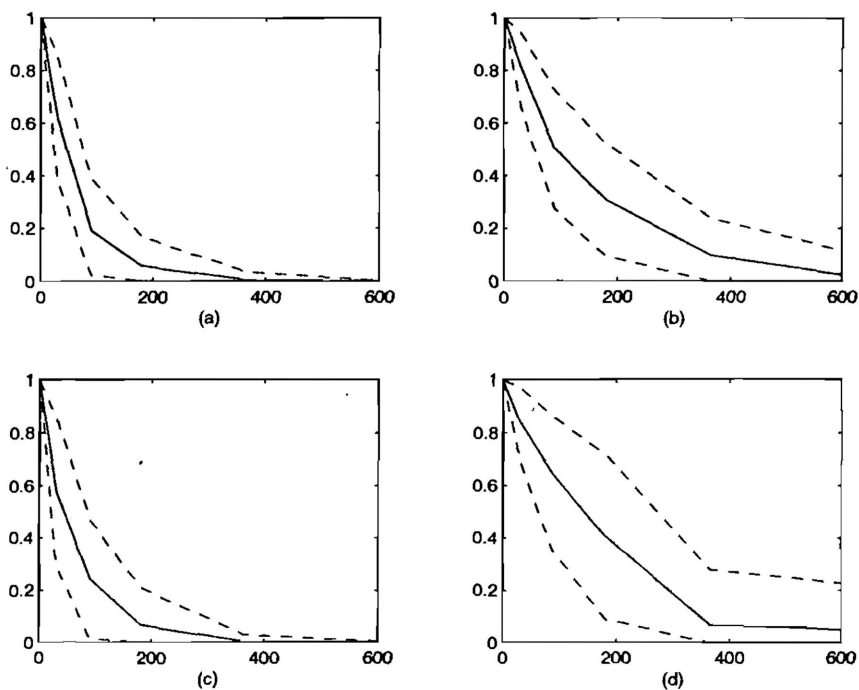


Figure 4. Estimated conditional survival functions of second failure time at age 50, given $T_1 < 30$ days, with 95% HPD intervals. Clayton model in (a) and (b) and conditional model in (c) and (d) for men and women respectively.

8. Summary

In this paper two models are proposed to generalize the model of Guo and Lin (1994) and Goggins and Finkelstein (2000) by introducing conditional dependence for bivariate failure times, and by using a Bayesian analysis. This makes it possible to distinguish between the effects of the covariates on the two failure times. The conditional model specifies interval probabilities for the first failure type as well as probabilities for the second failure type, conditional on the first failure. This works well in the bivariate case when the number of intervals is not too large, but a disadvantage is that the number of parameters increases exponentially if we want to extend the model to more than two dimensions. The Clayton model also works well, can be easily extended and will have a manageable number of parameters in the multivariate case. However, the single measure of association in the model is too restrictive to model adequately the relationships between more than two variables. If parametric models are assumed for the marginal distributions, a number of copulas are available for modelling the data and extension to more than two dimensions should follow more easily.

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Appendix

Likelihood for the conditional bivariate model

Consider the third term in the product on the right-hand side of equation (3.2). It gives the likelihood corresponding to an item whose first event was censored in the j th interval while the second event was a failure during the q th interval. The probability is

$$\begin{aligned}
 P[T_1 > t_{1j}, t_{2,q-1} < T_2 < t_{2q}] &= P[T_2 \in I_q | T_1 > t_{1j}] P[T_1 > t_{1j}] \\
 &= \sum_{s=j+1}^{r+1} P[T_2 \in I_q | T_1 \in I_s] \\
 &\quad \times P[T_1 \in I_s | T_1 > t_{1j}] P[T_1 > t_{1j}] \\
 &= \sum_{s=j+1}^{r+1} P[T_2 \in I_q | T_1 \in I_s] \\
 &\quad \times P[T_1 \in I_s, T_1 > t_{1j}] \\
 &= \sum_{s=j+1}^{r+1} P[T_2 \in I_q | T_1 \in I_s] \\
 &\quad \times P[T_1 \in I_s] \quad \text{for } s > j \\
 &= \sum_{s=j+1}^{r+1} (1 - P_{2qs}) \prod_{\ell=1}^{q-1} P_{21s} (1 - P_{1s}) \prod_{\ell=1}^{s-1} P_{1\ell}.
 \end{aligned}$$

The other terms of (3.2) follow similarly.

Simulating from the bivariate Clayton model with Weibull marginals

The marginal densities are $T_m \sim \text{Weib}(\alpha_m, \phi_m)$, $m = 1, 2$, and with the Cox proportional hazard model and covariate $\mathbf{z}_i$, the survival function of T_{1i} , $i = 1, 2, \dots, n$, is given by

$$S(t_{1i} | \mathbf{z}_i) = \exp\left(-(\lambda_{1i} t_{1i})^{\phi_1}\right),$$

where $\lambda_{mi} = \alpha_m \exp(\beta_m \mathbf{z}_i / \phi_m)$, $m = 1, 2$.

The covariates $\mathbf{z}_i$, $i = 1, 2, \dots, n$ are drawn from a $N(0, 1)$ distribution and then

$$t_{1i} = \lambda_{1i}^{-1} (-\log(u_{1i})), \quad i = 1, 2, \dots, n,$$

where $U \sim \text{Uniform}(0, 1)$.

The joint CDF of T_1 and T_2 is

$$\begin{aligned} F(t_{1i}, t_{2i}) &= 1 + S(t_{1i}, t_{2i}) - S(t_{1i}) - S(t_{2i}) \\ &= 1 + (\exp(\omega_{1i}) + \exp(\omega_{2i}) - 1)^{-\theta} - \exp(\omega_{1i}) - \exp(\omega_{2i}), \end{aligned}$$

where

$$\omega_{mi} = \theta^{-1} (\lambda_{mi} t_{mi})^{\phi_m}, \quad m = 1, 2.$$

The conditional CDF of T_2 , given $T_1 = t_{1i}$, is given by

$$F(t_{2i} | t_{1i}) = 1 - \exp((\theta + 1) \omega_{1i} (\exp(\omega_{1i}) + \exp(\omega_{2i}) - 1)^{-(\theta+1)}).$$

The conditional value of t_{2i} is then obtained as

$$t_{2i} = \frac{\theta}{\lambda_{21}} \left[\log \left\{ 1 - \exp(\omega_{1i}) \left(1 - (1 - u_{2i})^{-1/(\theta+1)} \right) \right\} \right]^{1/\phi_2}.$$

Besides the simulated values censored at the end of the sixth interval, 20% of the remaining values were randomly selected and censored by multiplying by a $\text{Uniform}(0, 1)$ random variable. The observations were then grouped into six intervals for each of the two failure types.

Markov chain Monte Carlo algorithm

Consider the Clayton model. Starting values $\beta^0, \gamma_1^0, \gamma_2^0$ and θ^0 are used to evaluate $f(\beta^0 | \gamma_1^0, \gamma_2^0, \theta^0)$. Using a random walk chain, a value β^* is drawn from the proposal density $N(\beta^0, \Sigma_1)$. The new value is accepted with probability

$$\rho = \min \left\{ \frac{f(\beta^* | \gamma_1^0, \gamma_2^0, \theta^0)}{f(\beta^0 | \gamma_1^0, \gamma_2^0, \theta^0)}, 1 \right\}, \text{ so that } \beta^{(1)} = \begin{cases} \beta^* & \text{with probability } \rho \\ \beta^0 & \text{with probability } 1-\rho. \end{cases}$$

In similar fashion $\gamma_1^{(1)}, \gamma_2^{(1)}$ and $\theta^{(1)}$ are obtained to complete one cycle. The proposal density for γ_m^* is $N(\gamma_m^0, \Sigma_m)$, $m = 1, 2$, while the proposal density for θ^* is $\text{Uniform}(0.6\theta^0, 1.4\theta^0)$. Similar proposal densities are used for the conditional model. The covariance matrices Σ_1 and Σ_m are diagonal with elements chosen so that the acceptance ratio is between 40% and 50%. For both the illustration in section 6 and the application in section 7, 20 000 cycles were run after burn-in to obtain samples from the marginal distributions of all parameters.